

Research paper

Flow field analysis and performance evaluation of a mini-hydrocyclone for ultra-low inlet flow rate

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ARTICLE INFO

Keywords:

Mini-hydrocyclone
Oil-water separation
Separation performance
Migration trajectory

ABSTRACT

It is difficult for hydrocyclone to separate oil-water under the condition of ultra-low inlet flow rate, which commonly exist in the field of energy engineering. Mini-hydrocyclone is an effective technology to solve this problem. The flow field characteristics and oil droplet migration trajectory in the mini-hydrocyclone are the key to guide its efficient work. An innovative mini-hydrocyclone are studied via computational fluid dynamics (CFD) and experiment method. The good separation performance of mini-hydrocyclone for ultra-low inlet flow rate and tiny particle size of oil droplets is verified and analyzed. Furthermore, The distribution of velocity, oil volume fraction and oil migration trajectories in the mini-hydrocyclone are analyzed systematically. The effects of oil droplet size, inlet flow rate and split ratio on the separation performance of mini-hydrocyclone are investigated. The results show that the maximum separation efficiency of the mini-hydrocyclone for 1.0 L/min ultra-low inlet flow rate is 99.98 %, when the split ratio is 30 %, the oil phase volume fraction is 2 %. Moreover, the morphological and distribution of oil core in mini-hydrocyclone are basically consistent between simulation and experiment results, under the conditions of different operating parameters. The minimum error of numerical simulation and experiment results is less than 2.23 %, the accuracy of the numerical simulation method is verified.

1. Introduction

The hydrocyclone (Liu et al., 2023) is a device based on centrifugal separation principles, utilizing density differences between substances to effect the separation of multiphase media. Due to its uncomplicated structure, exceptional performance, and efficient classification capabilities, hydrocyclones find extensive application across various domains such as heterogeneous medium separation (Chang et al., 2021), imicrobial collection (Liu et al., 2019), chemical removal (Zheng et al., 2024), food purification (Yang et al., 2013) and biotechnology (Ji et al., 2023). Currently, hydrocyclone has become one of the main technologies of oil-water separation in energy field. However, it is difficult to achieve high efficiency separation under the condition of ultra-low inlet flow rate and tiny oil droplets due to the lack of sufficient fluid swirl strength and radial gradient force for separation (Li et al., 2023a). Therefore, efficiently separating tiny oil droplet has become an important problem in the field of energy engineering.

Previous theoretical and experimental studies by researchers show that the separation accuracy of small particle size discrete media

improves as the main diameter of the hydrocyclone decreases (Lv et al., 2020; Li et al., 2016). Therefore, the design and research of small hydrocyclones which can realize the efficient separation of fine particles under the condition of ultra-low inlet flow rate are gradually rising. They analyze the performance of the mini-hydrocyclone in terms of structural parameters and operating parameters. Yuan et al. (2020) explored the influence of overflow structure parameters on the separation performance of hydrocyclones with diameter 20 mm by single factor method. It is concluded that when the diameter of overflow tube is 4.5 mm and the depth of overflow tube is 10 mm, the micro hydrocyclone has the highest degree of purification of materials. Cilliers et al. (2018) utilized hydrocyclone with diameter of 10 mm, by changing floc size and feed concentration, to test the effect of flocculation degree on the separation performance of hydrocyclone, concluded that the separation performance of hydrocyclone is significantly improved with the increase of flocs, which shows that the micro-hydrocyclone has a good application potential in the pre-concentration of low concentration imicrobial suspensions. Wu et al. (2018) used hydrocyclone with a diameter of 8 mm to analyze the effects of starch concentration, split ratio and

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feed rate on the adsorption efficiency and separation efficiency of micro hydrocyclone by response surface method, indicating that starch concentration, split ratio and feed flow rate have influence on adsorption efficiency and separation efficiency, and the order of influence on adsorption efficiency is starch concentration > split ratio > feed flow rate. Maira et al. (2020) adopted a method combining 3D printing with soft lithography to construct a 3 mm transparent mini-hydrocyclone on a polydimethylsiloxane block for the swirling separation of mixture samples of stem cells and microcarriers in the medical field. The results show a 95 % separation efficiency for samples containing a mixture of stem cells and microcarriers. Mognon et al. (2015) designed a miniature hydrocyclone with a diameter of 30 mm for the field of solid-liquid separation in the petroleum industry, and optimized the parameters such as the length and height of the vortex detector, the truncated cone angle and the downstream diameter, after optimization, the separation efficiency is kept above 89 %. Xing et al. (2022) utilized the response surface optimization design method to optimize the structural parameters of the oil-water separation hydrocyclone under ultra-low feed rates, establishing the quadratic effect relationship between structural parameters and separation performance. In conclusion, scholars have made significant advancements in the structural design of mini-hydrocyclones, laying an important theoretical foundation for further research on micro-hydrocyclones.

In addition, other researchers promote the separation of medium with small particle size by adding external field. Mofarrah et al. (2019) designed an enhanced cyclone separator using high voltage electric field, improved the separation performance of pulverized coal from 88.3 % to 96.2 % by electric field enhancement. Zhang et al. (Zhang et al., 2018). used magnetic field enhancement technology to combine temperature field, magnetic field, and flow field to study and analyze the fractionation escape rate of submicron particles with different magnetic flux densities, indicating that the magnetic confinement effect can effectively reduce submicron particle escape in an electrostatic cyclone precipitator. On the separation of oil phase medium in micro hydrocyclone, Zhang et al. (2023) carried out a study on the ultra-clean separation of particulate pollutants from micro-hydrocyclones with different overflow diameters, verified the reduction law of short flow in convective field in different overflow microchannels. In conclusion, researchers have conducted studies on the separation of micro-particles in the mini-hydrocyclone, laying a theoretical foundation. However, there is limited research on the separation process of tiny oil droplets in the mini hydrocyclone and the analysis of the flow field in the mini-hydrocyclone. At present, there is little clear understanding of the influence of the migration process of small oil droplets in the mini-hydrocyclone on the separation performance.

In summary, the above researches on mini-hydrocyclone focuses on the design of new structure, parameter optimization and applicability to different media. The adaptability of mini-hydrocyclone for ultra-low inlet flowrate and fine discrete media has been verified. Otherwise, aforementioned literatures have laid a theoretical foundation for the proposal of new structure hydrocyclone suitable for ultra-low inlet flow rate. However, there are few reports on the distribution characteristics of flow field and the migration trajectory of oil droplets in mini-hydrocyclone, which are not only the theoretical basis to analysis its separation performance, but also the key to guide equipment improvement and parameter optimization. In this paper, on the basis of the mini-hydrocyclone proposed by our earlier literature, the flow field characteristics and oil droplet migration trajectories of the mini-hydrocyclone are analyzed by numerical simulation method. The effects of overflow split ratio, inlet flow rate and main diameter on the separation performance of mini-hydrocyclone are investigated systematically. The results of this research not only provide theoretical reference for the innovative design of hydrocyclones but also supply technical support for optimal parameters control and engineering applications of mini-hydrocyclones.

2. Structure and method

2.1. Structure and working principle of mini-hydrocyclone

The mini-hydrocyclone structure is shown in Fig. 1. The investigation into various inlet structures of the conventional mini-hydrocyclone reveals that the pressure field and velocity field within the mini-hydrocyclone with a double inlet structure exhibit favorable symmetry and stability. Moreover, the mini-hydrocyclone featuring a double tangential inlet also demonstrates superior separation efficiency (Li et al., 2023b; Dong et al., 2021). Therefore, the double tangential inlets structure is designed to improve the stability of flowfield in mini-hydrocyclone. In addition double cone segment structure is designed to realize the lifting of oil droplet in mini-hydrocyclone. During separation, oil water mixture enter to mini-hydrocyclone through double tangential inlets and rotates at a high speed, the oil phase gathers in the central area of the mini-hydrocyclone and moves upward due to the smaller density, eventually flow out from overflow, the water rotates along the mini-hydrocyclone wall and flow out from the underflow.

It can be seen that the entire mini-hydrocyclone structure is divided into four sections: cylindrical section, large cone section, small cone section, and underflow section. It is important to analyzing the flow field in different sections of hydrocyclone for understanding the flow field characteristics in the mini-hydrocyclone (Fang et al., 2021; Zhang et al., 2022; Gonçalves et al., 2022; Li et al., 2023c). Therefore, five cross-sectional positions are specified to qualitatively analyze the flow field characteristics of mini-hydrocyclone. These analysis cross-sections are denoted as $Z_1=6$ mm, $Z_2=8$ mm, $Z_3=13$ mm, $Z_4=35$ mm and $Z_5=60$ mm, as shown in Fig. 1.

Five types of mini-hydrocyclones with different demention levels are designed to further verify the advantages of mini-hydrocyclone miniaturized design for ultra-low liquid inlet flow rate and tiny particle size oil droplets. Proportional amplification of size parameters on the basis of our previously proposed mini-hydrocyclone is conducted and the magnification factor is 1.5, 2.0, 2.5 and 3.0, respectively. The numerical simulation and separation experiment are carried out to compare the separation performance of different mini-hydrocyclones under the condition of various ultra-low inlet flow rate and tiny oil droplet. The structural specifications for each scaling factor are shown in Table 1.

2.2. Grid generation and grid independence verification

Hexahedral meshes are employed due to their high stability and good convergence compared to other meshes (Wang et al., 2018; Peng et al., 2023). In the optimization process, the mini-oil-water separation mini-hydrocyclone model utilized hexahedral grids as the fundamental

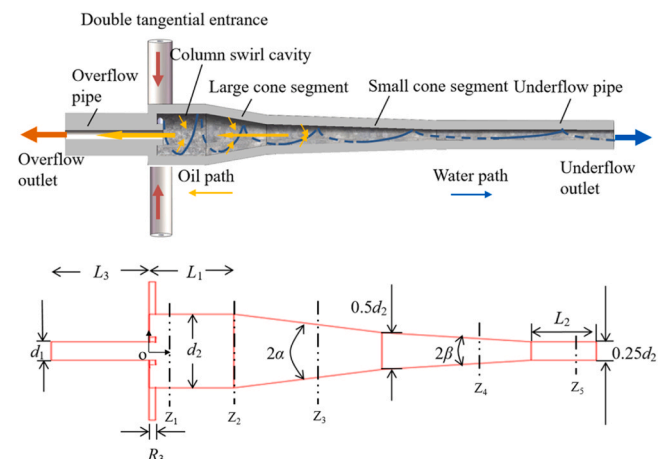


Fig. 1. Structure and working principle of mini-hydrocyclone.

Table1
Initial model structure parameter table.

Factors	Diameter of overflow outlet d_1 (mm)	Main diameter d_2 (mm)	Length of Cylindrical L_1 (mm)	Length of underflo w outlet L_2 (mm)	Length overflow Outlet L_3 (mm)	Overflow insertion depth R_3 (mm)	Angle of large-cone swirl chamber α ($^\circ$)	Angle of small-cone swirl chamber β ($^\circ$)
Model 1	1.5	7	8	29	15	1	10	1.8
Model 2	2.25	10.5	12	43.5	22.5	1.5	10	1.8
Model 3	3.0	14	16	58	30	2	10	1.8
Model 4	3.75	17.5	20	72.5	37.5	2.5	10	1.8
Model 5	4.5	21	24	87	45	3	10	1.8

mesh unit. Theoretically, a higher grid count enhances calculation accuracy (Li et al., 2021; Guang et al., 2013). Fig. 2 shows the schematic diagram of the model meshing. The aspect ratio serves as an indicator to evaluate grid quality, where the value closer to 1 signifies better grid quality. It is worth noting that the grid's aspect ratio is maintained above 0.5. However, an excessive number of grids isn't advantageous for computations. Once the grid count surpasses a certain point, it has marginal effects on the structure. For the initial structural model, this paper tested six different grid parameters as shown in table 2. Level-1=337, 382, Level-2=428, 997, Level-3=496, 081, Level-4=553, 944, Level-5=648, 918, and Level-6=728, 591 to evaluate grid independence, using tangential velocity as the criterion as shown in Fig. 3. The computation time for each grid level is as follows: Level-1 is 10 hours, Level-2 is 13 hours, Level-3 is 15 hours, Level-4 is 17 hours, Level-5 is 21 hours, and Level-6 is 24 hours. The results indicated that once the grid number exceeded 496, 081, the variation deviation between tangential velocity is less than 3 %. Considering the model size and the grid independence test results, the model with 496,081 grid number is chosen for the study to mitigate the influence of grid count on simulation outcomes. The influence of the number of grids on computing cost is mainly reflected in the tradeoff between computing accuracy and computing resource requirements. Increasing the number of grids can improve the calculation accuracy and accurately capture the details of droplet dynamics and heterogeneous medium separation, but at the same time significantly increase the calculation time and resource consumption. Reducing the number of meshes can reduce the

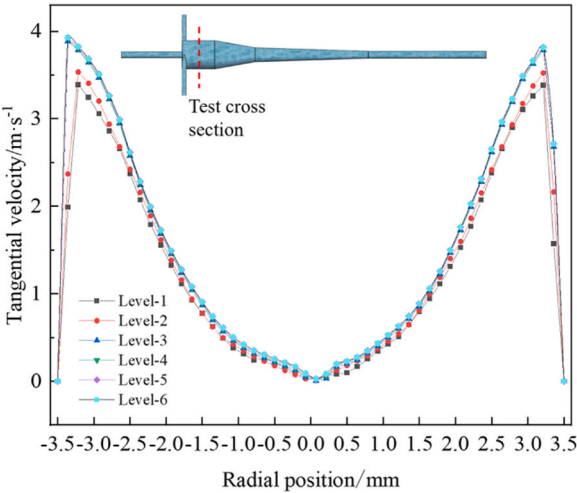


Fig. 3. Grid independence test.

computational cost, but it may lose accuracy and can not fully simulate complex physical phenomena. Therefore, when selecting the number of grids, it is necessary to comprehensively consider the required accuracy and available computing resources in order to find the best balance.

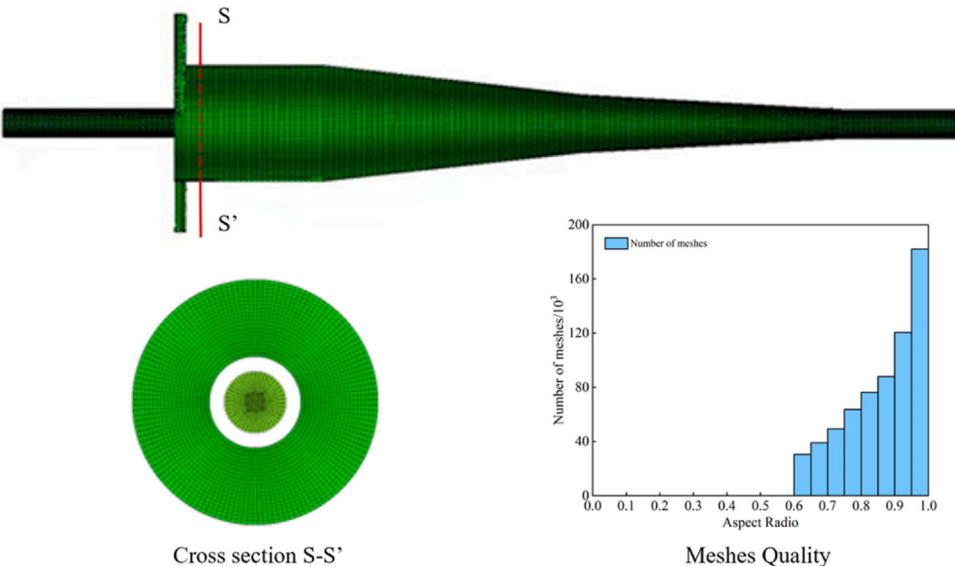


Fig. 2. Model meshing diagram.

Table 2

Details of different grid levels.

Level	Number of grids	Computation time (hours)
1	337,382	10
2	428,997	13
3	496,081	15
4	553,944	17
5	648,918	21
6	728,691	24

2.3. Boundary condition and mathematic model

ANSYS-fluent, a commercial software to simulate fluid flow is utilized to simulate the separation performance and flow field characteristics of mini-hydrocyclones. The three-dimensional models of the fluid domain of mini-hydrocyclone are created by Solidworks software, then converted into a *x_t* format file, and then imported into ANSYS meshing software for grid generation. Finally, the grid model of mini-hydrocyclone is imported into ANSYS-fluent software for analysis. The Mixture multiphase flow model is employed to simulate the oil and water flow in mini-hydrocyclone. The continuous phase is the water with the density is 997 kg/m^3 and the viscosity is $8.94 \times 10^{-4} \text{ Pa}\cdot\text{s}$ at 25°C . The oil phase is oil field crude oil, with the viscosity is $3.46 \times 10^{-3} \text{ Pa}\cdot\text{s}$ at 25°C , measured using a Malvern rotational rheometer, and a density is 868 kg/m^3 , measured using a balance. In the conventional hydrocyclone separation process, the diameter of oil droplets at the underflow outlet measures less than $100 \mu\text{m}$, as measured by the Malvern 2000 particle size meter. The turbulence calculation model utilized the RSM (Reynolds Stress Model), with a turbulence intensity of 0.047. The inlet was set as a velocity inlet, and both the overflow and underflow outlets were set as free outlets. The simulation aimed for a convergence accuracy of 10^{-6} .

The Reynolds number is a significant dimensionless parameter in fluid mechanics, employed to characterize the relative significance of inertia force and viscous force during fluid motion. Its definition is as follows:

$$Re = \frac{\rho v D}{\mu} \quad (1)$$

Where ρ is density of fluid (kg/m^3), D is equivalent diameter of pipe (m) v is the velocity of the fluid (m/s).

The inlet flow rate Q can be expressed as

$$Q = v \cdot s \quad (2)$$

Where s is the area of inlet (m^2).

It can be observed that the velocity of the fluid v is associated with both the inlet flow rate Q and the Reynolds number. Therefore, the inlet flow rate is used to express the performance analysis under the condition of different liquid inlet velocity.

In the process of numerical simulation, the continuity equations of water and oil are as follows:

$$\frac{\partial}{\partial t}(\alpha_c \rho_c) + \nabla \cdot (\alpha_c \rho_c u_c) = 0 \quad (3)$$

$$\frac{\partial}{\partial t}(\alpha_d \rho_d) + \nabla \cdot (\alpha_d \rho_d u_d) = 0 \quad (4)$$

The equation represents a mass conservation equation describing a two-phase flow system, where α represents volume fraction, ρ represents density (kg/m^3), and u represents the velocity of each phase (m/s). In the equations: α_c is the volume fraction of continuous phase (water); ρ_c is the density of continuous phase (oil); u_c is the velocity of continuous phase (water, m/s); α_d is the volume fraction of discrete phase (oil); ρ_d is the density of discrete phase (oil, kg/m^3); u_d is the velocity of discrete phase (oil, m/s).

In the mini-hydrocyclone, continuity equations for water and oil phases play a key role in describing the motion behavior and velocity distribution. These equations, also termed mass conservation equations, serve several essential functions: Continuity equations characterize the continuous flow of fluid mass within the mini-hydrocyclone in both space and time, unveiling the intricacies of fluid motion. They allow for the determination of velocity distributions within the mini-hydrocyclone's flow field, including both magnitude and direction. This is crucial for understanding how the fluid moves within the hydrocyclone and how different phases behave. The continuity equations ensure that mass is conserved throughout the fluid flow process within the mini-hydrocyclone. Particularly in multiphase flow scenarios, such as oil and water mixtures, these equations ensure accurate mass transfer and distribution between the phases. These equations also provide the necessary boundary and initial conditions for simulations within the mini-hydrocyclone. These conditions are vital for initiating and maintaining the simulation process accurately and stably (Liu and Fu, 2023).

The essence of RSM model is based on the law of time homogenization to construct the transport equation of Reynolds stress directly (Wang et al., 2017; Wang and Liu, 2004), the newly generated unknown terms in the transport equation are simplified respectively, thus a system of closed equations is formed. The choice of the Reynolds Stress Model (RSM) over other turbulence models like k-epsilon, k-omega, Spalart-Allmaras (S-A), or Shear Stress Transport (SST) models in Computational Fluid Dynamics (CFD) simulations is often based on the specific characteristics of the flow being studied and the objectives of the analysis. RSM is often preferred when dealing with flows that exhibit complex turbulence characteristics, such as swirling flows, separated flows, or flows with significant anisotropic turbulence effects. RSM provides a more comprehensive description of turbulence compared to simpler models like k-epsilon or k-omega.

For incompressible flow, the Navier-Stokes equation after time homogenization is:

$$\frac{\partial u_i}{\partial x_i} = 0 \quad (5)$$

$$\rho \frac{\partial u_i}{\partial t} + \rho \frac{\partial (u_i u_j)}{\partial x_j} = - \frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left(\mu \frac{\partial u_i}{\partial x_j} - \rho \overline{u_i u_j} \right) \quad (6)$$

In which, x_i and x_j are coordinate components; u_i and u_j are time homogenization velocity components; p is time homogenization pressure, Pa; μ is the hydrodynamic viscosity, Pa·s; ρ is the fluid density, kg/m^3 ; $\overline{u_i u_j}$ is the Reynolds stress component.

In the RSM model, the above basic equations are closed by solving the transport equations of each component of Reynolds stress:

$$\frac{\partial (\rho \overline{u_i u_j})}{\partial t} + \frac{\partial (\rho u_k \overline{u_i u_j})}{\partial x_k} = D_{ij} + P_{ij} + \varphi_{ij} + \varepsilon_{ij} \quad (7)$$

In which, diffusion term D_{ij} , Stress generation term P_{ij} , Pressure strain term φ_{ij} and viscous dissipative term ε_{ij} are respectively:

$$D_{ij} = \frac{\partial}{\partial x_k} \left(\mu_t \frac{\partial \overline{u_i u_j}}{\partial x_k} + \mu \frac{\partial \overline{u_i u_j}}{\partial x_k} \right) \quad (8)$$

$$P_{ij} = -p \left(\overline{u_i u_j} \frac{\partial u_j}{\partial x_k} + \overline{u_j u_k} \frac{\partial u_i}{\partial x_k} \right) \quad (9)$$

$$\varphi_{ij} = -C_1 \rho \frac{\varepsilon}{k} \left(\overline{u_i u_j} - \frac{2}{3} \delta_{ij} k \right) - C_2 \left(P_{ij} - \frac{1}{3} \delta_{ij} P_{kk} \right) \quad (10)$$

$$\varepsilon_{ij} = \frac{2}{3} \rho \varepsilon \delta_{ij} \quad (11)$$

Where, μ_t is turbulent viscosity; C_1 is 1.8; k is turbulent kinetic energy; ε is the dissipation rate; δ_{ij} is a Kronecker-delta function, $P_{kk}=2p$.

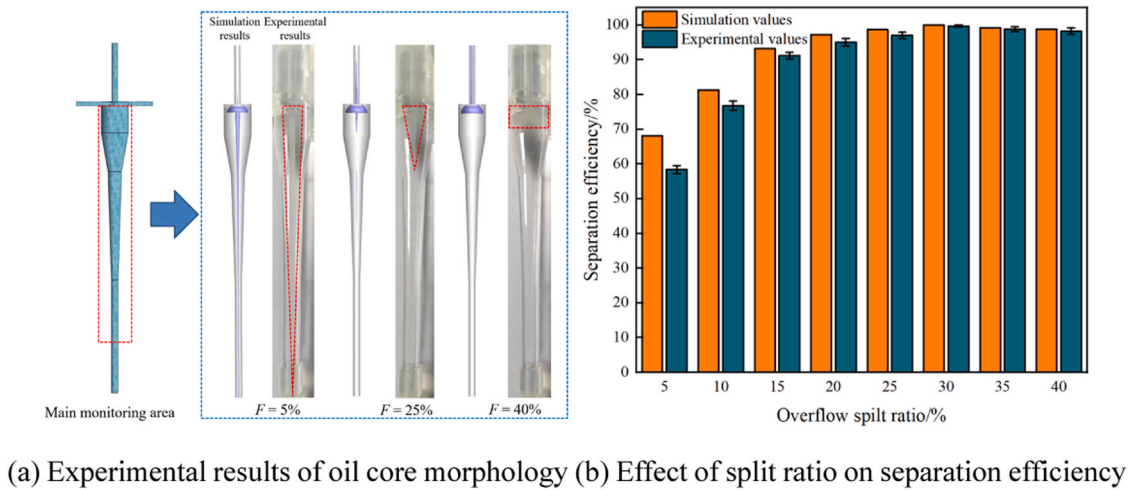


Fig. 5. Comparison of experimental and simulated values under different split ratios.

analyzed the average error of experimental efficiency and simulation efficiency of hydrocyclone by the same method, and the error was 3.45 %. In conclusion, the experimental values in this paper are in good agreement with the simulated values.

$$\Delta = \sum_{i=1}^n \frac{e_i - t_i}{nt_i} \times 100\% \quad (11)$$

Where, e is the predicted value of the model, t is the actual value obtained by simulation, i is the additional test number, and n is the number of additional test groups.

3. Results and discussion

3.1. The effect of main diameters on separation efficiency

The main diameter of mini-hydrocyclone determines its processing capacity and the size of separable oil droplets (Yablonskii, 2019; Fang et al., 2022). The suitability of the main diameter parameters for ultra-low inlet flow rates and tiny oil droplet size values is verified through the separation performance of hydrocyclones with different main diameter parameters and oil droplet size values. Fig. 6 shows the separation efficiency comparison of experimental and simulated values under with different main diameter hydrocyclones. The centrifugal force

of oil droplets decreases progressively with the main diameter parameter increases at ultra-low inlet flow rate, This restrictive the oil droplets to converge at the axis center. The increase of the main diameter parameter from 7 mm to 21 mm reduced the separation efficiency by 31.6 % when the oil droplet size is 300 μ m. As the size of the oil droplets decreases, the inertial force by the oil droplet in the hydrocyclone decreases. That results in a lack of energy for oil droplet transport to the centre and a significant reduction in separation efficiency. However, the separation performance of the hydrocyclone with a main diameter of 7 mm is still higher than 84 % at an oil droplet diameter of 100 μ m. We processed an experimental hydrocyclone for testing, during which oil droplet diameter of 300 μ m (measured by Malvern particle size meter) is used, and the experiments and simulations presented a good agreement. The aforementioned analysis demonstrates that hydrocyclones with a 7 mm main diameter exhibit superior adaptability to tiny size of oil droplets and ultra-low inlet flow rates. Therefore, the main focus is on the mini-hydrocyclone with 7 mm diameter.

3.2. The effect of split ratio on separation efficiency

The separation efficiency under various split ratios is investigated using the mini-hydrocyclone with main diameter of 7 mm. Fig. 7 shows the variation of the hydrocyclone's separation efficiency under different split ratios. It's apparent that the separation efficiency increases

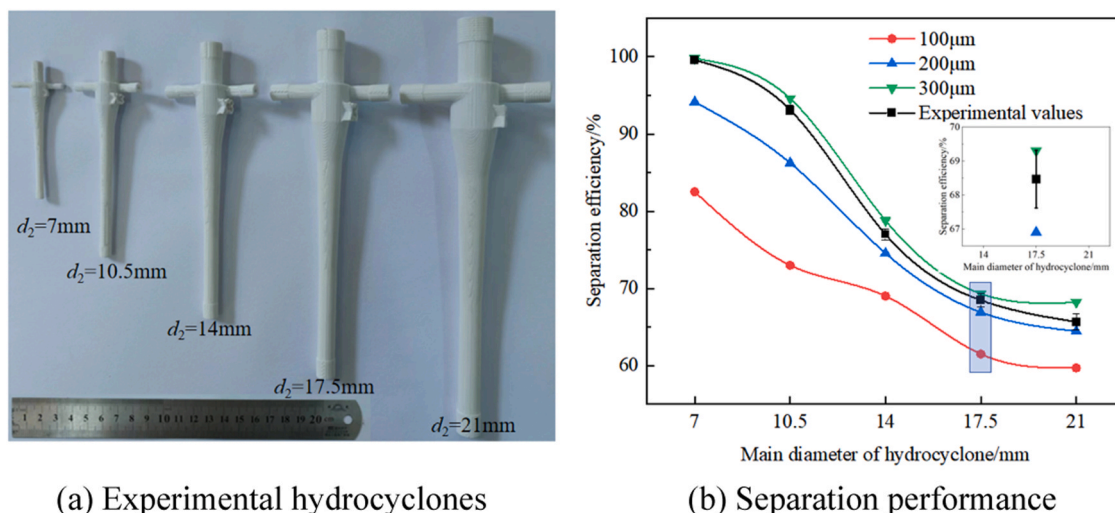


Fig. 6. Comparison of experimental and simulated values under hydrocyclones with different main diameter.

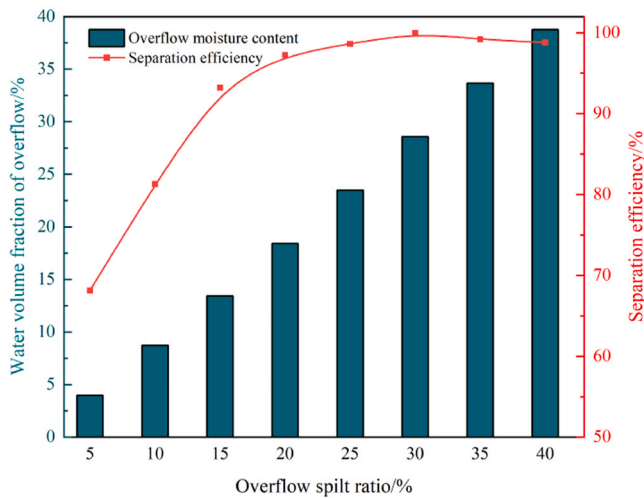


Fig. 7. Effect of separation performance under the condition of different different split ratios.

gradually with the increase of split ratio. As the split ratio increases from 4 % to 40 %, separation efficiency increased from 68.5 % to 99.98 %, and remains relatively stable with the further increase of the split ratio. Moreover, five cross sections (Z_1 – Z_5) of the mini-hydrocyclone are selected for more comprehensive analysis of the distribution of oil volume fraction in the mini-hydrocyclone as presented in Fig. 8. It can be observed that the oil volume fraction in the mini-hydrocyclone decreases gradually with increase of the split ratio. The smaller split ratio, the more severe the accumulation of the oil phase, leading to a higher oil phase volume fraction at each cross section. Conversely, the larger the split ratio, the smaller the oil phase volume fraction at each cross section. When $F = 40$ %, the oil phase volume fraction of the Z_4 and Z_5 sections approaches 0. The changing trend of separation efficiency with the split ratio is consistent with the research results of Xing et al (Xing et al., 2022). Therefore, this enhance of separation efficiency can be attributed that the width of the oil core reduces and the upward migration rate of the oil phase accelerates with the increase of split ratio.

3.3. The effect of inlet oil concentration on separation efficiency

The verification of separation performance under different oil concentrations is shown in Fig. 9. Despite a slight decrease in separation efficiency with increasing inlet oil concentration, the overall separation efficiency remains above 98.6 %. The trend is consistent with the research results of Li et al. (2023a). Meanwhile, compared with the mini-hydrocyclone structure of Li et al., the structure of this study shows a 10 % increase in separation efficiency under conditions of higher oil concentration. (5 %–10 %).

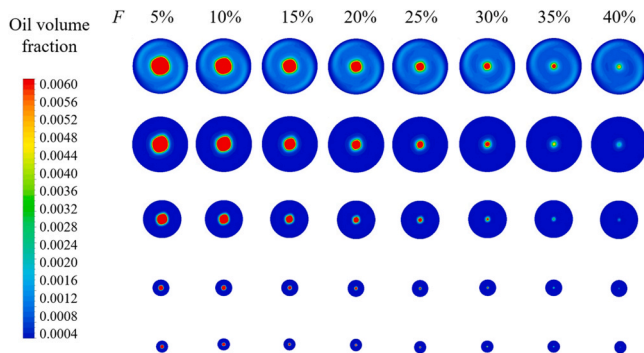


Fig. 8. Distribution of oil volume distribution in mini-hydrocyclone under different split ratios.

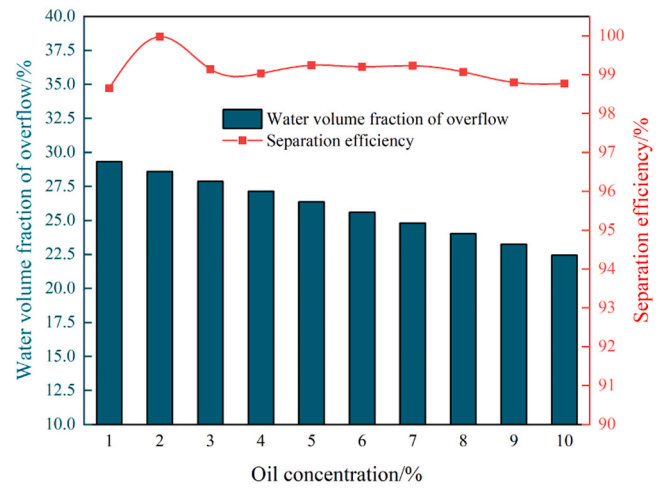


Fig. 9. Effect of separation performance under the condition of different inlet oil concentration.

Additionally, the overflow water phase volume fraction tends to decrease as inlet oil concentration increases. This phenomenon is attributed to the increased inlet oil concentration, which causes more oil phase to accumulate towards the overflow outlet, displacing some water medium from the area near the overflow. The inlet oil concentration contour of the tangential section under varying inlet oil concentrations is shown in Fig. 10. It illustrates that with increasing inlet oil concentration, the flow rate of the oil in the overflow rises. The oil within the swirl cavity expands, and the oil core gradually widens. Moreover, the residence time of certain oil phase media in the swirl cavity prolongs. It is obvious that the accumulation phenomenon is significant in the oil phase. When the oil concentration is 10 %, the oil core extends to the underflow outlet, which is attributed to the decreased separation performance.

Fig. 11 shows the migration trajectory and velocity of the oil under different inlet oil concentration. As the oil concentration rises, the residence time of the oil in the swirl cavity increases, extending the oil droplet escape process. Consequently, the migration trajectory of oil droplets gradually shifts downward. It can be observe from Fig. 11 that the residence time and trajectory of oil droplets increase with the increase of oil concentration, which explains the reason for the growth of oil core in Fig. 10.

3.4. The effect of inlet flow rate on separation efficiency

An analysis of the mini-hydrocyclone's performance under varying

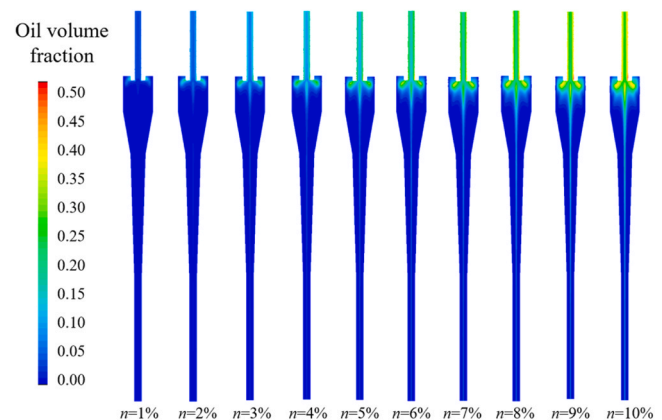


Fig. 10. Distribution of oil volume fraction in mini-hydrocyclone under the condition of different inlet oil concentration.

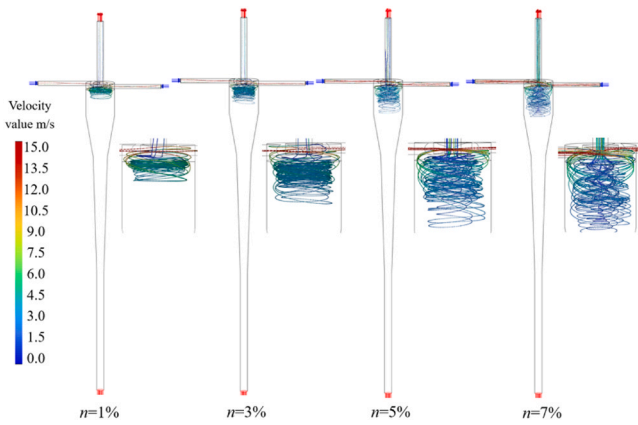


Fig. 11. Migration trajectories of oil droplets under different inlet oil concentration.

inlet flow rates is conducted, and the change diagram of separation performance is shown in Fig. 12. As the inlet flow rate increases, the separation performance initially improves, stabilizes within a certain range, but the swirl pressure drop continues to rise. The increase in pressure drop is indicative of an increase in energy consumption, which should be avoided. Therefore, when the inlet flow rate is 1.0 L/min, the pressure drop and efficiency reach the optimal balance point. To further analyze the tangential velocity under different inlet flow rates, the Z₃ and Z₄ sections of the large cone section and small cone section are selected.

The tangential velocity is the velocity component with the largest relative value and the greatest influence on the separation process in the mini-hydrocyclone. It is directly affected by the inlet velocity. Generally speaking, the greater the inlet velocity, the greater the tangential velocity. The tangential velocity of the small cone segment decreases at a relatively slower pace. At the same time, through the above research, it is found that the separation process of the mini-hydrocyclone mainly occurs in the large segment and the small cone segment. Therefore, the analysis of tangential velocity in the large cone section Z₃ and the small cone section Z₄ is shown in Fig. 13. It can be seen that the axial velocity increases at first and then decreases from the side wall to the axis. The figure shows that with the increase in the inlet flow rate, the tangential velocity within the mini-hydrocyclone rises, generating a stronger centrifugal force for mini-hydrocyclone separation. This enhancement promotes improved separation performance. The velocity varies across different positions, with a faster decrease in velocity observed in the

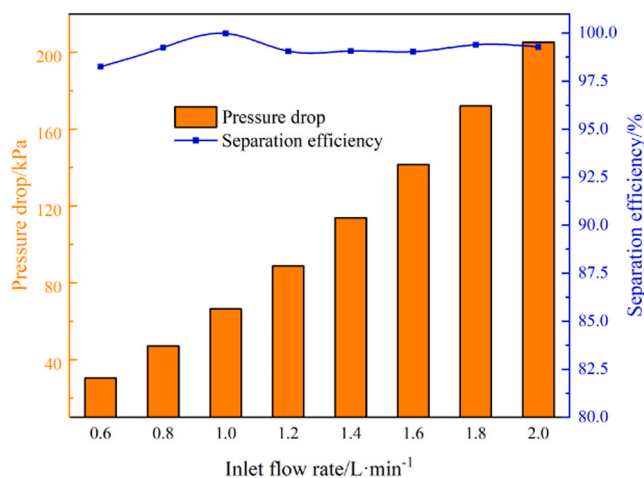


Fig. 12. Effect of separation performance under the condition of different inlet flow rate.

large cone section compared to the small cone section.

Fig. 14 shows the trajectories and velocities of oil droplets under varying inlet flow rates. With increasing inlet flow rate, while the overall particle trajectory shifts downward slightly, the migration distance of oil droplets within the swirl field increases. However, the swirl velocity in the flow field notably intensifies, leading to oil-free particles flowing from the bottom flow port. This affirms the separation performance of the mini-hydrocyclone under varying inlet flow rates.

The centrifugal force F_a received by the oil droplet in the swirling field is:

$$F_a = m_o a = \frac{\pi}{6} x^3 \rho_o \frac{v_t^2}{r} \quad (14)$$

In the formula, m_o is mass of oil droplet particles, kg; ρ_o density of oil droplet particles, kg/m³; r is oil droplet size, m; v_t is radial distance of oil droplet position from axis, m/s.

The equation illustrates that the greater the tangential velocity, the stronger the centripetal force acting on the oil droplet, making it easier to move the droplet towards the axis. This explains why increasing the inlet flow rate enhances the separation efficiency and facilitates the escape of oil droplets from the overflow outlet.

In conclusion, with the increase of inlet flow rate, the tangential velocity increases, and the increase of tangential velocity increases the centripetal force of oil droplets, which promotes the oil droplets to enter LZVV (Locus of Zero Vertical Velocity) ahead of time, improving the separation efficiency.

3.5. The effect of oil droplet size on separation efficiency

A numerical simulation study are carried out to determine the separation performance of mini-hydrocyclones for oil droplets of different sizes, while keeping the operating parameters constant. The range of oil droplet size parameters is set from 50 μm to 400 μm with a horizontal interval of 50 μm. Effect of separation performance under the condition of different sizes of oil droplet is shown in Fig. 15. It is observed that as the size of oil droplet increases, the centrifugal force acting on the oil droplet in the mini-hydrocyclone also increases, leading to a significant improvement in overall separation efficiency and a slight decrease in overflow water volume fraction. This is consistent with the research results of Liu et al. (Liu et al., 2022). The separation efficiency of the mini-hydrocyclone is 59.0 % when the oil droplet size is 50 μm. It is shown that its separation cut size is under 50 μm according to the definition of separation particle size. In addition, the separation efficiency of the mini-hydrocyclone is maintained above 95 % when the oil droplet size exceeds 150 μm, which verifies the high separation accuracy of the mini-hydrocyclone. Compared with the hydrocyclone performance of Liu et al., It has more advantages in the separation of oil droplets with small particle size less than 150 μm.

Fig. 16 shows distribution of oil volume fraction in mini-hydrocyclone under different size of oil droplets. When the oil droplet size is tiny, the distribution of the oil in the mini-hydrocyclone is relatively dispersed, and the oil droplets do not receive enough energy to converge to the axis center. According to Eq. 14, as the size of oil droplet increases, the energy provided by the centrifugal force increases, facilitating the entry of oil droplets into the LZVV (Locus of Zero Vertical Velocity) of the mini-hydrocyclone, and thus flowing out from the overflow outlet. Fig. 17 shows the migration trajectories of oil droplets under different oil droplet sizes. It is evident that as the size of oil droplet increases, the downward migration distance of oil droplets in the mini-hydrocyclone also increases, and they tend to move around the overflow outlet. This indicates that the increase in the size of oil droplets leads to a decrease in the axial position of oil droplets entering LZVV, bringing them closer to the overflow outlet. This observation is confirmed by the oil distribution depicted in Fig. 16. Comparing Fig. 16 and Fig. 17, when the oil droplet size is 50 μm, the oil droplet trajectory is distributed in the whole hydrocyclone, and the oil core is also extended to the underflow

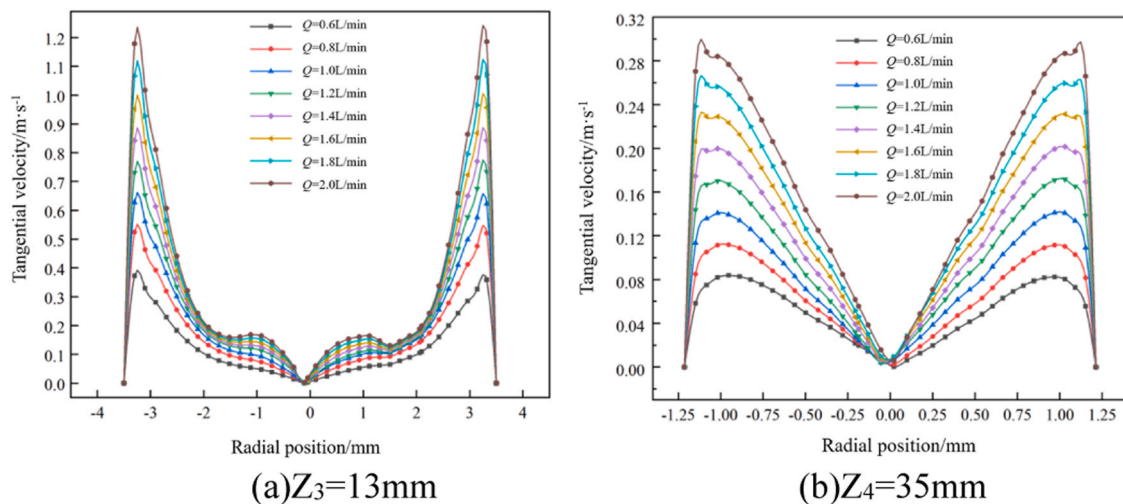


Fig. 13. Tangential velocity distribution at different cross sections.

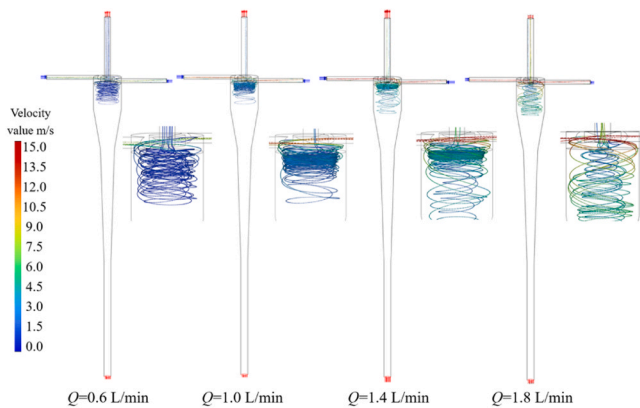


Fig. 14. Migration trajectories of oil droplets under different inlet flow rates.

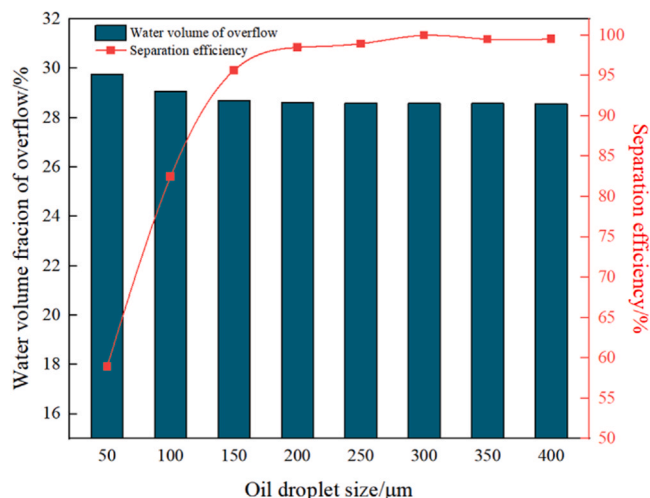


Fig. 15. Effect of separation performance under the condition of different oil droplet sizes.

outlet. When the oil droplet size is 350 μm , the trajectory of the oil droplet is mainly concentrated at the overflow outlet of the cylindrical section, and the oil core is also concentrated at the overflow outlet position.

4. Conclusion

In this paper, a new type of mini-hydrocyclone is studied using computational fluid dynamics and experiment methods. The high efficiency of the mini-hydrocyclone in handling ultra-low inlet flow rates and tiny oil droplet sizes is verified and analyzed. Meanwhile, the velocity distribution, oil content, and migration trajectory in the mini-hydrocyclone are systematically analyzed. The conclusions drawn from this study are as follows:

1. The analysis of mini-hydrocyclones with varying diameters, demonstrate that smaller-diameter hydrocyclones exhibit superior adaptability to low inlet flow rates. When the liquid intake is 1 L/min, the overflow shunt ratio is 30 %, the oil concentration is 2 %, and the droplet size is 300 μm . At equivalent inlet flow rates, the separation efficiency of the 7 mm diameter mini-hydrocyclone exceeds that of the 21 mm mini-hydrocyclone by 41 %. Additionally, for oil droplet sizes of 100 μm , the separation efficiency of the 7 mm diameter mini-hydrocyclone is 84 %.

2. Analysis of the 7 mm diameter hydrocyclone was conducted using a single-factor design method to investigate its separation performance under different operational and physical parameters. Our findings indicate that optimal separation performance, reaching 99.98 %, is achieved when the overflow split ratio is set at 30 %, inlet oil concentration is 2 %, inlet flow rate is 1 L/min, and oil droplet size is 300 μm . Oil droplet particle injection is facilitated using the Discrete Phase Model (DPM). Analysis of particle trajectories further validated the oil phase separation performance of the mini-hydrocyclone across varied operational and physical parameters. When the oil droplet size exceeds 150 μm , the mini-hydrocyclone's performance remains above 95 %.

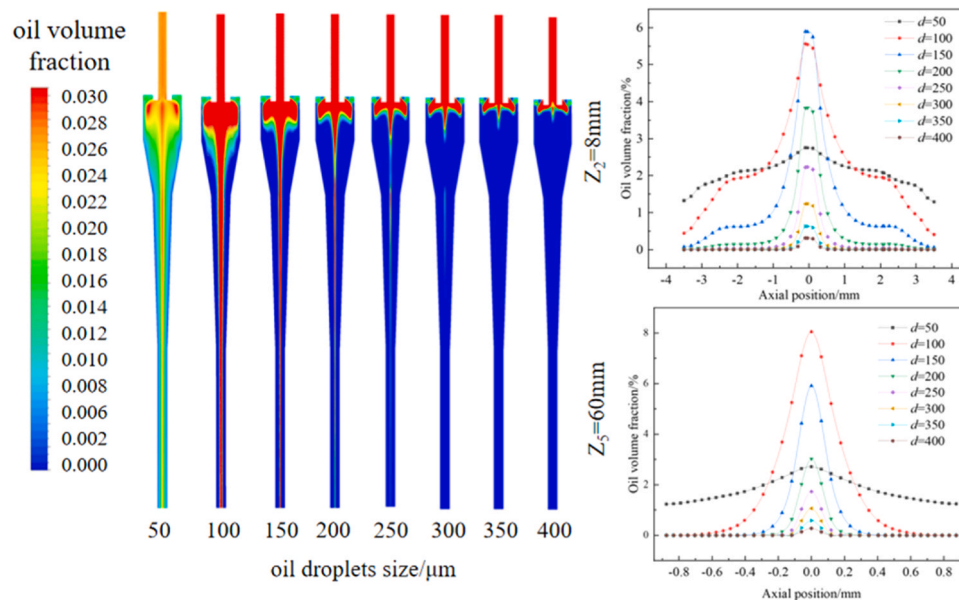
Through this study, the flow field analysis and performance evaluation of a miniature hydrocyclone under the conditions of ultra-low inlet flow rate and micro-oil droplet separation were carried out. However, in the process of oil droplet migration, it is worth exploring because the oil droplet itself will deform and exhibit some interesting dynamic behavior. The future will focus on the research in this direction.

CRediT authorship contribution statement

Meng Cai: Writing – review & editing, Writing – original draft, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence



(a) Distribution of oil volume fraction (b) Comparison of oil volume fraction

Fig. 16. Distribution of oil volume fraction in mini-hydrocyclone under different size of oil droplets.

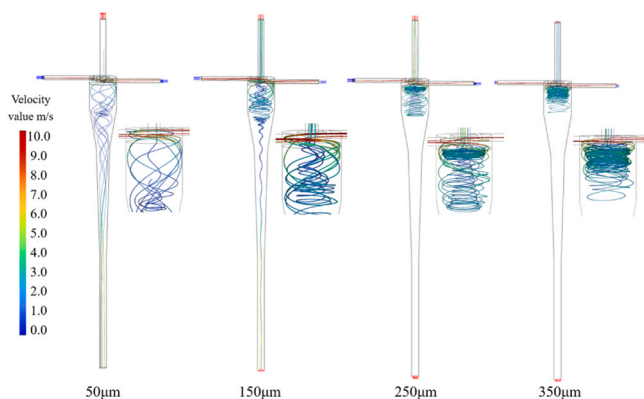


Fig. 17. Migration trajectories of oil droplets under different oil droplet sizes.

the work reported in this paper.

Data Availability

No data was used for the research described in the article.

Acknowledgments

This work was supported by The Key projects supported by Regional Innovation and Development Joint Fund of National Natural Science Foundation of China [Grant number U21A20104].

Author statement

Throughout the development of this paper, Meng Cai was responsible for the formulation of the research question, design of the study, data collection, and analysis.

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